

3D Coordinates Extension — Build the Snow Golem (Beginner)

Topic: 3D Snow Golem (x, y, z) · **Course:** 3D Coordinates · **Level:** Beginner (Extension, after Lesson 6) ·

Time: about 30 minutes

Build a standing snow golem in 3D — **snow body**, **pumpkin head**, **coal buttons**, **stick arms**. Every block needs **three** numbers (**x, y, z**): x **across**, y **up**, z **deeper** (forward).

- Stand on your **home spot** (red block) every run. Move, and the golem moves too.
- 🍷 Snow = **snow block**. Head = **carved pumpkin**. Buttons + eyes = **coal block**. Arms = a **brown block** (oak log).

1 · Predict 🧙

Read the code. Circle one word.

```
place snow block at (3, 1, 1)
place snow block at (3, 2, 1)
place snow block at (3, 3, 1)
```

Only y changes. This snow tower is going... Circle one: across · up · deeper

```
place snow block at (3, 3, 1)
place snow block at (3, 3, 2)
place snow block at (3, 3, 3)
```

Only z changes. Circle one: across · up · deeper

2 · Spot the Bug 🐛

A coal button sits **on the front** of the body at (3, 4, 1). The code below has only **two** numbers, so it lands flat on the ground.

```
# broken  
place coal block at (3, 4)
```

Fill the blank to fix it:

```
place coal block at (3, 4, __)
```

Which number fills the blank? Circle one: 1 · 3

What was missing? Circle one: the z number · the color

3 · Show Your Work 📷 🎥

Look at the 🧊 snow golem in the world. Walk around it. See the pumpkin head up top, the coal buttons down the front, and the two arms poking out the sides.

Build a small snow golem on your home spot. Keep it easy:

Snow body: fill snow blocks from (2, 1, 1) up to (4, 5, 1) — a tall tower.

Pumpkin head on top: fill carved pumpkin from (2, 6, 1) to (4, 7, 1).

Two coal eyes on the head: coal block at (2, 7, 1) and (4, 7, 1).

Two coal buttons on the body: coal block at (3, 4, 1) and (3, 2, 1).

Two arms out the sides: a brown block at (1, 4, 1) and (5, 4, 1).

💡 Tip: the head goes **up** (bigger y). The arms poke **out** (x = 1 and x = 5). Everything shows on the **front** (z = 1).

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera at the golem. Name the parts.

Say these out loud, filling in the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was ____.

Something new I learned was ____.

Submit ✅

Send this worksheet + your **one** video to teacher on KakaoTalk.